Java

Portable applications with classes and a managed runtime

THE MENTAL MODEL

Write a recipe in a standard format; a runtime follows it on many systems.

SOURCE → BYTECODE → RUNTIME → RESULT

READ THE CODE

```
class Main {
  public static void main(String[] args) {
    int[] xs = {10,20,30};
    int sum = 0;
    for (int x : xs) sum += x;
    System.out.println(sum);
  }
}
```

FOUR CLUES TO READ IT

- class groups code around a named type.
- main is the example entry method.
- for (int x : xs) visits each array element.
- System.out.println prints one line.

WHAT MAKES IT DIFFERENT

- Usually executes via the Java Virtual Machine (JVM), a managed runtime.
- Garbage collection reclaims many unused objects automatically.

WHERE IT FITS

Enterprise systems, backend services, Android legacy ecosystems.

WATCH OUT

Portable does not mean every platform setup or library is identical.

TRY ALONE • 60 SEC

Change the array to {3,4,5}; predict the printed sum.